AETHERFIN GPU OPS

GenAI GPU FinOps & Carbon Sustainability Platform

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Portfolio Deliverable for Enterprise Product Analyst & Data Engineer Reviews

Executive Summary

Core Value Proposition: Bridges the visibility gap between raw machine load telemetry and business-level finance/environmental indexes.

Total Compute Spend: Realized a baseline operational spend of $7,844.98 across 5 active cluster nodes.

GICP Cost Waste Penalty: Identified $5,073.76 (64.7%) in wasted spend on idle instances running under 15% load.

Compute Efficiency: Serving average of 2.23 tokens per Wh, benchmarked across model deployments.

Environmental Impact: Tracked 467.32 kg in Scope 3 carbon footprint emissions.

Problem Statement

Skyrocketing Infrastructure Budgets: Generative AI model training and serving represent the largest modern enterprise IT expense.

The Visibility Vacuum: Standard server tools show CPU/Memory load but omit cost, SLA penalties, and carbon metrics.

Severe Resource Waste: Multi-million dollar GPU clusters remain reserved 24/7 with zero load outside working hours.

Regulatory Reporting Deficits: Compliance audits require tracking Scope 3 green IT emissions, but standard cloud metrics fail to correlate electricity grid coefficients.

Business Context

Financial Leakage: Up to 60% of GPU compute budgets are wasted on idle allocations.

Regulatory Compliance: Imminent Scope 3 environmental carbon footprint audits are legally required in global jurisdictions.

The Green IT Opportunity: Optimizing compute sizing and regional scheduling yields double benefits: lowering spend and carbon footprint.

Stakeholder Alignment: Designed to support Chief Financial Officers (CFOs), VP of Engineering, AI Product Managers, and Sustainability Officers.

Product Vision

Continuous Intelligence: Convert low-level machine logs into high-level business analytics indexes.

Actionable Sizing Engine: Feed recommendation pipelines directly to container orchestration systems to shut down waste.

Risk-Balanced Simulation: Allow product managers to simulate cost and SLA performance changes before allocating clusters.

Immutable Audit Trails: Guarantee security compliance through JWT auth and immutable transaction auditing logs.

System Architecture

Multi-Tier Design: Decoupled client, REST API server, and relational database tiers.

Frontend Client: React TypeScript application executing custom ECharts rendering inside a premium Glassmorphism Zinc dark UI.

FastAPI Backend: High-performance Python ASGI API routing, using NumPy for sliding-window mathematical Z-scores.

Database Layer: SQLAlchemy ORM layer abstracting SQLite (development) and MySQL (production) servers.

Orchestration Stacks: Dockerized compose profiles managing application services, database engines, and pgAdmin monitors.

Database Design & Schema

gpu_cluster_nodes: Node-level registry tracking provider (AWS/GCP/On-Prem), region, instance type, hourly cost, and grid carbon intensity.

model_deployments: Active serving configurations mapping allocated GPUs, latency SLAs, and target TPS thresholds.

gpu_utilization_logs: Telemetry logging hourly average GPU loads and actual power draw in Watts.

carbon_emissions_logs: Correlating power draws to regional grid intensity to log emissions hourly.

inference_request_logs: Transaction logging query parameters, token numbers, latency, and SLA violation status.

Technology Stack

Frontend: React 18, TypeScript, Vite, ECharts-for-React, Lucide icons, and custom CSS.

Backend: Python 3.11+, FastAPI, Uvicorn, SQLAlchemy, PyJWT, Cryptography, and NumPy.

Database: MySQL (Local/Production), SQLite (Testing), PyMySQL, and PostgreSQL support.

Deployment & DevOps: Docker, Docker Compose, Git repository source control, and Vercel hosting integration.

Innovative KPI Framework

GPU Idle Cost Penalty (GICP): Translates technical load metrics into dollar waste assessments.

Model Compute Efficiency Score (MCES): Tracks token output density relative to physical energy consumption.

SLA Violation Exposure Index (SLA-VEI): Measures user latency breaches, weighting extreme delays non-linearly.

Carbon Offset ROI (COROI): Correlates financial savings of regional rerouting actions against offset expenses.

GPU Idle Cost Penalty (GICP)

Formula: GICP = sum(Idle Hours * Hourly Cost) / Total Spend * 100

Threshold Definition: Tracks hours where a node is active but average GPU load stays under 15%.

Business Cost: In the audited trial, $5,073.76 out of $7,844.98 was wasted on idle capacity.

Outcome Action: Identifies exact instances suitable for decommissioning or dynamic scale-down.

Model Compute Efficiency Score (MCES)

Formula: MCES = (Prompt Tokens + Completion Tokens) / sum(Power Draw in Watts * 1 Hour)

Standardization Value: Allows C-suite comparison of model architectures on different hardware types.

Hardware Insights: Llama-3-8B on GCP L4 nodes achieved 8.42 tokens/Wh, compared to 1.12 tokens/Wh for Llama-3-70B on H100s.

Outcome Action: Identifies candidate models for right-sizing or quantization.

SLA Violation Exposure Index (SLA-VEI)

Formula: SLA-VEI = sum(max(0, (Actual Latency - Target Latency) / Target Latency)) / Total Requests * 100

The Severity Weighting: Counts of violations do not capture user frustration; SLA-VEI weights delays exponentially.

Baseline Score: Current baseline score is 2.89, driven by peak diurnal traffic queuing delays.

Outcome Action: Signals when cost-cutting measures threaten customer retention.

Carbon Offset ROI (COROI)

Index Concept: Evaluates the cost efficiency of shifting workloads to low-intensity regional grids.

Grid Differences: Compares On-Premises coal grid (650g CO_2/kWh) to GCP Europe-West1 nuclear grid (45g CO_2/kWh).

The Trade-Off: Carbon routing may increase network latencies but completely eliminates carbon penalty offsets.

Outcome Action: Scheduler routes batch jobs to clean regions during off-peak times.

Executive & Dashboard Previews

Dashboard Interface (sizing.png): KPI overview widgets, active node grid, and 48-hour utilization timelines.

Outlier Analytics (anomalies.png): Z-score outlier graphs detailing historical server spikes.

KPI Drilldowns (drilldown.png): Sliding sidebar panels displaying node-level cost allocations.

Security & Auditing Portal (admin_logs.png): Immutable transaction log tables displaying JWT user activity.

Trend & Cohort Analytics

Multi-Range Filters: Supports 7-day, 30-day, and custom timeline aggregation windows.

Diurnal Load Profiles: Identifies node demand spikes (peak hours 9 AM - 6 PM) to plan scheduling.

Cohort Analysis: Compares model execution profiles to pinpoint underperforming instances.

Green Savings Tracking: Measures GICP improvements and emission reductions post-optimization.

Z-Score Anomaly Detection

Dynamic Baselines: Avoids static thresholds by computing sliding-window mean and standard deviations.

Latency Spikes: Flags transactions exceeding a Z-score of 2.2 as high severity spikes.

Idle Waste Warnings: Triggers alerts for nodes running under 10% average load for 12+ consecutive hours.

Root Cause Diagnostics: Automatically pairs anomalies with technical diagnoses and routing solutions.

AI FinOps Advisor

Context Integration: Feeds active cluster costs, idle penalties, and carbon metrics directly to the LLM context.

Natural Language Interface: Answering queries like 'How can I reduce costs by 20%?' or 'Which nodes are underperforming?'.

Actionable Recommendations: Provides consulting-grade suggestions detailing weekly cost savings.

Technical Fallback Rules: Rule-based fallback executes on the database if the LLM API is offline.

What-If Sizing Simulator

Predictive Sizing Controls: Adjust scale multiplier, daily operating hours, and traffic concurrency.

Routing Strategies: Toggle default, Cost-Optimized (cheapest), and Carbon-Optimized (cleanest) modes.

Trade-Off Modeling: Simulates how cost-cutting changes affect SLA violation risks.

Real-Time Previews: Computes projected costs, carbon footprints, and net savings dynamically.

Performance & Load Audits

Optimization Focus: Resolved DB query loops by pre-fetching deployment records into memory dictionaries.

100 Concurrency: 100% Success | P50 latency 804.1ms | Peak RAM 93.9 MB

500 Concurrency: 100% Success | P50 latency 886.2ms | Peak RAM 95.3 MB

1,000 Concurrency: 100% Success | P50 latency 967.6ms | Peak RAM 96.9 MB

Security & Compliance Audits

Access Protection: Bearer JWT validation shields all secure API routes and telemetry reports.

Type Safety Validation: Pydantic schemas filter and reject malformed JSON payloads (HTTP 422).

SQL Injection Prevention: Parameterized queries in SQLAlchemy ORM eliminate injection vectors.

CORS Access Controls: Restricted origins block unauthorized external domains from querying backend API endpoints.

Business Impact Summary

Double-Bottom Line Value: Shows that optimizing compute resources reduces spend and carbon emissions simultaneously.

Cost Savings: Cut weekly cluster run-rates by 50% through smart sizing.

Carbon Reduction: Achieved a 73% drop in Scope 3 greenhouse gas footprints.

Enterprise Readiness: Audit trails and RBAC controls satisfy compliance audits.

Cost Optimization Actions

Terminating Idle Node: Decommissioning `aws-us-east-1-a100-idle` saves $2,284.80 / week.

Off-Peak Auto-Scaling: Downsizing H100 counts by 50% between 11 PM and 6 AM saves $478.24 / week.

Model Right-Sizing: Migrating Mixtral from GCP A100 to GCP L4 instances saves $806.40 / week.

Net Sizing Business Outcome: Delivers $3,921.58 / week ($16,980 monthly savings, 50.0% cost reduction).

Environmental Sustainability Outcomes

Grid Schedulers: Batch jobs are automatically routed to cleaner regional grids during off-peak hours.

Stable-Diffusion XL Migration: Shifting image batch jobs to GCP Europe West-1 wind grid saves 343 kg CO_2 / week.

Net Environmental Sizing: Lowers weekly emissions from 467.3 kg CO_2 to 124.2 kg CO_2 (73.4% reduction).

Offsets Optimization: Avoids $356 / month in carbon offsets, accelerating ESG timeline targets.

Future Roadmap

Autonomous Cloud Scaling: Connect the sizing API to AWS/GCP dynamic node groups to scale pools directly.

Predictive Load Forecasting: Implement regression models to forecast peak traffic and pre-allocate spot instances.

Direct Offset Marketplaces: Integrate carbon credit purchasing APIs directly into the management panel.

Multi-Tenant SaaS: Add organization isolation schemas to serve multiple enterprise accounts.

Conclusion

Recruiter-Ready Platform: Successfully bridges GenAI compute operations, financial auditing, and carbon ESG metrics.

Sophisticated Calculations: Implements custom formulas (GICP, MCES, SLA-VEI, COROI) on local telemetry.

Production-Grade Performance: Validated at 1,000 requests concurrency under 100% success rates.

Advanced Competency: Demonstrates full capability in Product Analytics, Data Engineering, and Security.